App Development 

Cadette Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors

The capstone “for good” badge

Today you design an app for **social good**:

- A real community problem
- A real user
- A defensible solution

The standard is professional: user research before design, design before code.

Apps for good, deconstructed 🔍

Look at two real social-good apps — a food-rescue app, a crisis-help finder. For each:

- Who's the **user**?
- What **problem**?
- **What's the smallest thing it does that helps?**

Meet the MVP

Minimum Viable Product.

Real developers ship the **minimum that solves the problem**, then grow it.

Big hearts. Small first versions.

User Research 🎤

Your team picks a community problem — a card, or better, a real one you know.

A mentor plays the user. Run a real interview:

- “Walk me through the problem.”
- “What have you tried?”
- “What would *solved* look like?”

Rule: no design talk until research is done ✋

Sketching during the interview means you've stopped listening.

Your deliverable 

[User] needs a way to [goal] because [barrier].

Plus the **top two must-haves** — in the user's own words.

User flow first 🗺️

Sticky-note map the user's path:

open app → do the core thing → confirmation
Three to five steps. No more.

Then wireframe 📐

1. One paper screen per sticky note
2. **The core action gets the biggest button**
3. **Accessibility pass** — big touch targets, readable text. Does it work for a user who is older, in a hurry, or new to phones?
4. **Feature-cut round** — anything not serving the problem statement goes

Accessibility is the “for good,” made concrete

An app that excludes users isn’t solving the community’s problem.

Cutting is designing.

Click Test 👆

The mentor-user thinks aloud while tapping your paper prototype.

Your team stays silent. Swap screens. Take notes.

Then make **one fix**.

The Pitch — 90 seconds ⏱️

1. The **problem**
2. The **user**
3. Your **MVP**
4. What you'd **build next** with more time

Optional: make it real 💻

In **MIT App Inventor** (appinventor.mit.edu):

Build the welcome screen + core button as a **running app** on a phone or emulator.

One working screen beats five imagined ones.

Robot moment 

We scope our robot each season to an **MVP** — the minimum machine that scores — before adding stretch features.

Same discipline. Bigger build.

What you accomplished ✅

- Explained what an MVP is and why developers start there
- Conducted user research and wrote a problem statement from it
- Designed a user flow and wireframes where every screen serves the user
- Ran a click test, iterated, and pitched the app’s purpose and next steps

Level up 🚀

- **appinventor.mit.edu** — free, and it builds real running apps
- Take your problem statement to the **actual person** and ask if you got it right
- Run the accessibility pass on an app you already use — what would you fix?

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This is how the program grows — and how the next troop and team get better materials.